

32. The activity of a radioactive sample is measured to be 18 MBq. What is its activity 3 half-lives before ?

- A. 6 MBq
- B. 54 MBq
- C. 72 MBq
- D. 144 MBq

31. A radioactive source contains a radioactive nuclide X which decays to become a stable nuclide Y . Besides X there is no other radioactive nuclides in this source. Which of the following statements **must be correct** after a time of one half-life of X has elapsed ?

- (1) The number of nuclides of X decreases to about half of its initial value.
- (2) The number of nuclides of Y increases to about twice of its initial value.
- (3) The activity of the radioactive source falls to about half of its initial value.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

*32. It takes 2.0 years for the activity of a sample containing a radioactive nuclide to drop from 800 Bq to 600 Bq. How long will it take for the activity of the sample to drop from 600 Bq to 300 Bq ?

- A. 2.0 years
- B. 2.5 years
- C. 3.0 years
- D. 4.8 years

33. Two radionuclides X and Y are of half-lives 3 hours and 4 hours respectively and initially there are N_X and N_Y undecayed nuclei respectively. After 24 hours, the number of undecayed nuclei of both nuclides becomes the same. Find $N_X : N_Y$.

- A. 8 : 1
- B. 4 : 3
- C. 4 : 1
- D. 2 : 1

35. A certain radioactive isotope X has a half-life of 20 hours. After a time interval of 10 hours, what is the approximate fraction (f) of a sample of the radioactive isotope X remaining?

A. $\frac{1}{4} \leq f \leq \frac{1}{2}$

B. $f = \frac{1}{2}$

C. $\frac{3}{4} > f > \frac{1}{2}$

D. $f > \frac{3}{4}$

*32. The half-lives of some radioisotopes are tabulated below.

radioisotope	half-life
carbon-11	20.3 minutes
phosphorus-32	14.3 days
sodium-22	2.60 years

Which of the following statements is/are correct ?

- (1) The activity of carbon-11 must be the highest.
- (2) The decay constant of phosphorus-32 is larger than that of carbon-11.
- (3) If the initial activity of sodium-22 is 1520 Bq, its activity would be lower than 380 Bq after 6 years.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

*33. A piece of wood found in an archaeological site is dated using carbon-14 dating method. It registers a corrected count rate of 11.0 counts per minute while a fresh wood sample cut from the same kind of trees gives a corrected count rate of 15.6 counts per minute. What is the approximate age of the wood found in the archaeological site ? Given: half-life of carbon-14 is 5730 years.

- A. 890 years
- B. 1300 years
- C. 2000 years
- D. 2900 years

31. A radioactive source undergoes decay such that 300 nuclides have decayed in 1 minute. Its activity is

A. 0.2 Bq.

B. 5 Bq.

C. 300 Bq.

D. 18000 Bq.

*35. Polonium-210 is a pure α -emitter with a half-life of 140 days and it will decay into lead, which is stable. Initially there is a sample containing 420 mg of pure polonium-210. Estimate the mass of polonium-210 left after 70 days.

- A. 315 mg
- B. 297 mg
- C. 210 mg
- D. 105 mg

33. In order to locate the cracks in an underground water pipe, a small amount of a radioisotope is added to the water in the pipe. The radioactive substance will leak through the crack to the soil nearby, thus radiation will be detected above ground. Which radioisotope (P , Q , R or S) below is the most suitable for this purpose ?

	isotope	half-life	radiation emitted
A.	P	66 hours	β
B.	Q	36 hours	γ
C.	R	7.2 hours	α
D.	S	45 minutes	α, β

*32. Radioactive sources X and Y have initial activities 100 kBq and 200 kBq respectively. One day later, their activities are found to be equal. After another day, the activity of X becomes 80 kBq. Deduce the corresponding activity of Y .

- A. 40 kBq
- B. 50 kBq
- C. 89 kBq
- D. 160 kBq

*32. The decay constant of a radioisotope of an element

- A. is random.
- B. depends on pressure and temperature.
- C. is directly proportional to the number of nucleons in the isotope.
- D. is an identifying characteristic of that isotope.

- *32. X and Y are two radioactive nuclides. The ratio of the mass of an atom of X to that of an atom of Y is $1 : 2$. The half-lives of X and Y are T and $2T$ respectively. If two samples consisting of purely X and Y respectively have the same initial mass, find the ratio of the number of undecayed nuclei of X to that of Y after a period of $4T$.

- A. $1 : 4$
- B. $1 : 2$
- C. $1 : 1$
- D. $2 : 1$

33. Two radioactive samples P and Q are freshly prepared. It is found that when $\frac{15}{16}$ of all the nuclei of P have decayed, $\frac{63}{64}$ of all the nuclei of Q have also decayed. Find the ratio $\frac{\text{half-life of } P}{\text{half-life of } Q}$.

- A. 1 : 4
- B. 2 : 3
- C. 3 : 2
- D. 4 : 1

32. Workers in nuclear power plants wear clothes with film badges to measure the dosage of radiation received over a period of time. Which type of radiation below **CANNOT** be monitored by the film badges ?

- A. α -radiation
- B. β -radiation
- C. γ -radiation
- D. X-rays

32. Which of the following units is used for measuring the radiation equivalent dose of a patient receiving radiotherapy treatment ?

- A. sievert (Sv)
- B. becquerel (Bq)
- C. joule (J)
- D. watt per square metre (W m^{-2})

*35. On which of the following does the activity of a radioactive source depend ?

- (1) the nature of the nuclear radiation emitted by the source
- (2) the half-life of the source
- (3) the number of active nuclides in the source

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only